

Kratik Rath

+1 (812) 822-7764 | Chicago, IL | rathi.kratik08@gmail.com | [linkedin.com/in/kratikrathi](https://www.linkedin.com/in/kratikrathi) | Portfolio: kratik-rathi.github.io | github.com/Kratik-Rathi

WORK EXPERIENCE

Data Scientist at Project 990 Inc. | Chicago, IL

October 2025 - Present

- Consolidated 15M+ IRS grant records spanning 7 years into a production database by engineering a Python ETL pipeline, eliminating duplicate ingestions and ensuring full audit traceability through exception logging and year-level summary reports.
- Built a dashboard using Plotly, React, and FastAPI, deploying on AWS EC2 to handle 2M+ records and reducing load time from 5 seconds to 90 ms.
- Architected an analytical data layer for 2M+ records powering the dashboard using DuckDB with B-tree indexing, server-side pagination and caching, reducing backend query response time to 30 ms.
- Delivered interactive Tableau dashboards visualizing 750K+ U.S. nonprofits by geographic distribution, enabling grant makers to drill down from national to city level and identify high-need regions by revenue density and nonprofit concentration.
- Processed and analyzed 4.5M IRS tax return records using Python, SQL and Alteryx to map nonprofits across the US and generate grant-making and grant-seeking recommendations.

Software Engineer Intern at Genmark AI Inc. | Remote, USA

July 2025 - August 2025

- Developed a RAG application using Claude API, Python and LangChain, embedding GCP Cloud Bucket data into a Milvus vector database and deploying via a Dockerized Streamlit interface, enabling 60% faster content generation for clients.
- Lowered content creation time to under 5 minutes by building a tool that analyzes the top 20 trending TikTok videos, generates transcripts via Whisper and produces personalized videos using Google Veo 2.

Research Assistant at Indiana University – Indiana Innocence Project | Bloomington, IN

January 2025 - May 2025

- Constructed a Python-based ingestion pipeline using BeautifulSoup to scrape and normalize legal case data from intake forms and judicial sources into a PostgreSQL database, cutting manual data entry time from 30 minutes to 1 minute.
- Designed a Looker Studio dashboard to visualize case timelines, severity and procedural status, helping assign review priority to 200+ cases.

Data Analyst Intern at Swastika Investmart Ltd. | Indore, India

January 2024 - July 2024

- Created an analytics pipeline using Python, Flask API, SQL, and cron jobs to analyze 50K+ product-level and campaign-level lead records across client zones, delivering daily reports to stakeholders and boosting conversions by 20% while cutting manual reporting by 90%.
- Engineered a scalable Docker-based scheduler to centralize internal API deployment and integrated Prometheus for unified logging, cutting debugging time by 60% and streamlining future rollouts for the team.

PROJECTS

California Live Wildfire Analysis Dashboard

December 2025

- Architected an ETL pipeline using AWS Lambda, S3, CloudWatch and Google BigQuery with an end-to-end runtime of 2.5 seconds, continuously ingesting and refreshing CAL FIRE wildfire incident data.
- Built a live Tableau wildfire analytics dashboard tracking incidents since 2017 across 58 California counties, with year-over-year KPIs, seasonal and duration breakdowns to identify high-impact regions and trend shifts.

IndyGo – Ridership Dashboard (Client Project, ENGR-E 583: Information Visualization) May 2025

- Ensured ridership accuracy by validating 38K GPS and APC records, eliminating 3% invalid entries through trip and route-specific thresholds.
- Designed an interactive Power BI dashboard using DAX calculations to surface weekday vs. weekend travel patterns, stop-level passenger flows and route-level ridership KPIs across 35K+ valid records, driving a 3% improvement in service planning.

DocuVerse – Chatbot

January 2025

- Developed a RAG-based document processing app with LangChain and Hugging Face embeddings to automate text chunking and semantic representation, reducing manual document review time by 80%.
- Enhanced retrieval with FAISS and integrated Gemma 2 to enable context-aware Q&A and structured summarization. Implemented persistent chat history and deployed the system on Streamlit for real-time usage.

EDUCATION

Indiana University – Bloomington

Master of Science in Data Science

August 2024 - May 2026

GPA: 3.6

Medi-Caps University

Bachelor of Technology in Computer Science and Engineering

August 2019 - July 2023

GPA: 3.4

TECHNICAL SKILLS

Programming Languages: Python, SQL, R

Databases & Warehousing: Snowflake, BigQuery, MySQL, PostgreSQL, MS SQL Server, MongoDB, DuckDB

Cloud Platforms & Big Data Technologies: Databricks, Apache Spark (PySpark, Spark SQL), AWS, GCP

ETL & Data Visualization: Alteryx, Power BI (DAX, Power Query), Tableau, Looker Studio, Excel

Libraries & Frameworks: Pandas, NumPy, Scikit-learn, PyTorch, TensorFlow, Keras, OpenCV, LangChain, Matplotlib, Seaborn, Streamlit, Flask, Airflow

Generative AI & LLMs: RAG, NLP, FAISS, Milvus, Whisper, Hugging Face

Machine Learning: Regression Models, Tree-based Models (XGBoost, Random Forest), SVM, Clustering (K-Means, GMM), Deep Learning (CNN), Feature Engineering, Statistical Modeling

Development Tools: Git (GitHub, GitLab), Docker, CI/CD, REST APIs, Postman, FastAPI, React

CERTIFICATIONS

AWS Certified Cloud Practitioner (January 2026)